

# Auditing AI Output

Data Can Deceive · Causal Roadmaps with (and without) AI

## Module E

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# A Confident Answer Is Not a Correct One

- The model will **not** tell you when it is wrong; fluency is not validity.
- “Human in the loop” does *not* mean reviewing the output and nodding.
- It means **auditing each claim** against the evidence that would justify it.

## Our test case

An **adjustment set** for a **target-trial protocol** [1] — the list of variables an analysis will control for.

**Objective:** turn a confident list of covariates into a set of checkable claims.

# Test Case

## Our prompt

*“Among adults initiating an SGLT2 inhibitor versus a DPP-4 inhibitor for type 2 diabetes, does the SGLT2 inhibitor reduce hospitalization for heart failure? List the variables to adjust for so the groups are comparable.”*

## The model's answer

*“Adjust for: age, sex, baseline eGFR, baseline HbA1c, prior heart failure, prior MI, baseline diuretic use, HbA1c at 6 months, and change in eGFR on treatment. Adjusting for glycemic control and renal response removes any residual imbalance in disease severity.”*

**Side note:** models vary and change fast, a good answer today can be a bad one tomorrow. The key point is to build the habit of auditing a model's answer *regardless* of how confident it sounds.

## Step 1: First Pass

### Answer in the chat

Read the nine variables suggested by the model. **Flag any you would question** and say why in a few words.

- age
- sex
- baseline eGFR
- baseline HbA1c
- prior heart failure
- prior MI
- baseline diuretic use
- HbA1c at 6 months
- change in eGFR on treatment

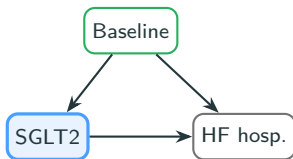
## Step 2: Audit

- Both time-invariant and baseline covariates measured *before* time zero are legitimate confounders: **ACCEPT**.
- HbA1c at 6 months and on-treatment eGFR change are measured **after** treatment starts.
- They sit on the causal pathway (**mediators**) and can act as **colliders**.
- Adjusting for them **blocks part of the effect** you want and can **manufacture bias**.

# Baseline vs Post-Baseline

## Baseline confounder

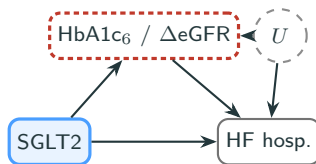
measured *before* time zero



**Adjust** to close the backdoor.

## Post-baseline variable

HbA1c<sub>6</sub> /  $\Delta$ eGFR, measured *after* time zero



**Do not adjust** — removes the *mediated* part of the total effect *and* opens collider bias via  $U$ .

**Assumption:** the collider path needs  $U \rightarrow M$  (unmeasured severity affects on-treatment response).

Drawing out the DAG helps make these assumptions clearer.

## Step 3: Decide and Justify

- Every field gets one of three verdicts: **ACCEPT** / **CORRECT** / **REJECT**.
- Each verdict is **paired with the evidence** that justifies it (otherwise it is just another confident assertion).
- For each covariate in your set, answer:
  - Was it measured **before time zero**?
  - **Where does it sit on the DAG** (confounder, mediator, collider)?

### Next step

**ACCEPT** keep as-is    **CORRECT** keep but fix (definition / timing / add a missing one)  
**REJECT** remove

## Step 4: Keep an Audit Log

| Variable                   | Timing        | Verdict | Justification                                                                          |
|----------------------------|---------------|---------|----------------------------------------------------------------------------------------|
| age, sex                   | pre-time-zero | ACCEPT  | Fixed at baseline; affect prescribing and HF risk.                                     |
| baseline eGFR, HbA1c       | pre-time-zero | ACCEPT  | Indication / severity confounders present before treatment.                            |
| prior HF, prior MI         | pre-time-zero | ACCEPT  | Strong confounders of treatment choice and outcome.                                    |
| baseline diuretic use      | pre-time-zero | CORRECT | Keep, but flag as a <i>proxy</i> for HF severity; define the lookback window.          |
| HbA1c at 6 months          | post-baseline | REJECT  | Mediator on $A \rightarrow Y$ ; collider with $U$ . Adjusting biases the total effect. |
| $\Delta$ eGFR on treatment | post-baseline | REJECT  | On the causal pathway; conditioning removes the mediated part and opens a collider.    |

# When the Mediator Is the Question

**ACCEPT** / **REJECT** assume you want the **total effect** of treatment. For that estimand, “post-baseline  $\Rightarrow$  reject” is right.

## Controlling for mediators

If the *mediated pathway itself* is your question, how much of the effect runs *through* HbA1c or eGFR, that is **mediation analysis**. With treatment-affected confounding it calls for **g-methods** (g-computation, IPW, TMLE) [2].

Reject the mediator *for a total effect*, and model it deliberately when it *is* the target. Do not apply the checklist bluntly.

# Your Turn!

**Target trial.** Among adults with COPD initiating **inhaled corticosteroids (ICS)** versus a long-acting bronchodilator alone, do ICS reduce **severe exacerbations** over the next year?

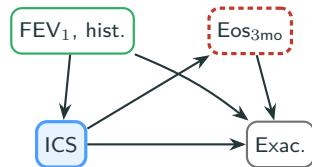
**Proposed adjustment set:**

- age
- smoking
- prior-year exacerbation count
- blood eosinophils at 3 months on ICS
- baseline blood eosinophils

**Your task:** for each variable, draw a DAG and run the audit test  $\Rightarrow$  **ACCEPT** / **CORRECT** / **REJECT**. Anything missing?

# Solution

- age, prior exacerbations, baseline eosinophils: **ACCEPT**  
pre-time-zero confounders of ICS use and exacerbations
- smoking: **CORRECT**  
keep, but define: baseline status / pack-years, before time zero
- eosinophils at 3 months: **REJECT**  
post-baseline mediator: ICS lowers eosinophils → fewer exacerbations
- missing* baseline FEV<sub>1</sub>: **CORRECT** (add)  
airflow severity confounds both ICS use and exacerbations



green = adjust   red box = do not

## Auditing any adjustment set

1. **Question the model's output:** is each variable measured *before* treatment starts? What is a given variable's position on a DAG (confounder, mediator, or collider)?
2. **Audit:** **ACCEPT** / **CORRECT** / **REJECT** for each. An omitted confounder is a finding too (**CORRECT** → add).
3. **Justify**
4. **Document**

The completed log *is* your causal assumptions made explicit and reproducible, as opposed to a black-box list of covariates [2].

**A confident answer is not a correct one.**

**Audit each claim against its evidence.**

## References

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- [1] Hernán MA, Robins JM. Using big data to emulate a target trial when a randomized trial is not available. *American Journal of Epidemiology*, 183(8):758–764, 2016.
- [2] Hernán MA, Robins JM. *Causal Inference: What If*. Chapman & Hall/CRC, 2020.